

Kyumin Lee

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EDUCATION

Chungbuk National University — B.S. in Software

Mar 2021 – Feb 2027 (expected)

School of Computer Science, College of Electrical and Information Engineering · Cheongju, South Korea

- **Cumulative GPA:** 3.95 / 4.50 (4th-year undergraduate)
- Admitted to the Dept. of Information & Communication Engineering (2021), then transferred to the School of Software.

Undergraduate Thesis / Research (in progress)

- **Working topic:** “Toward Reliable LLM-Based Agents: Incorporating User-Input and Environmental Constraints as Structural Priors.”
- Advised by Prof. Keon-Myung Lee (이건명), AI Laboratory, Chungbuk National University. Topic developed from hands-on analysis of agent harness internals and time-series modeling.

RESEARCH INTERESTS

- **LLM-based AI agents:** agent harness engineering, tool-use reasoning, and multi-agent orchestration.
- **Reliable and trustworthy autonomy:** human-in-the-loop control, evaluation of agent trajectories, and robustness to real-world input.
- **Time-series modeling:** forecasting and anomaly/fraud detection on temporal signals.
- **Applied NLP & on-device LLM systems.**

RESEARCH EXPERIENCES

Undergraduate Researcher — AI Lab, Chungbuk Nat'l University

Sep 2025 – Aug 2026

Advised by Prof. Keon-Myung Lee (이건명) · ongoing

- Time-series direction: studied TimeXer, a SOTA forecasting model, and ran it on financial time-series data.
- LLM-agent direction: analyzing the internals of an open-source AI agent and extending it into a multi-agent system (detailed under AI Projects below).

Research Trainee — ETRI

Jul 2025 – Aug 2025

Electronics and Telecommunications Research Institute, Digital Convergence Research Lab

Project: “Beyond X-verse” core technology for hyper-realistic physical–virtual interaction

- Built a unified viewer module in Python integrating motion-capture SDKs from three vendors — Xsens, OptiTrack (optical capture), and Manus (glove sensors).
- Synchronized multi-vendor sensor position values with real-time video and recorded/predicted subsequent motion; performed time-series processing of motion-sensor signals.
- Automated a manual conversion workflow that previously took 30+ minutes per case, removing a key bottleneck in the lab’s research cycle.

Field Trainee — KISDI

Dec 2024 – Jan 2025

Korea Information Society Development Institute, Budget Planning Team

- Proposed and built a Python program automating the generation of budget reports for KCA PMS and integrated Ezbaro submission, reducing repetitive administrative workload.

PROJECTS

Anti-Phishing AI Keyboard — LLM-based financial-fraud prevention

2025

Backend / AI Developer · Excellence Award, 12th National ICT Convergence Contest

- Designed a custom keyboard that streams typed text over WebSocket to a FastAPI backend, which performs real-time contextual analysis, scores financial-fraud risk in stages, warns the user, and notifies family members.
- Ran a local LLM (Gemma 3) for on-premise inference, and built a gating layer with the KIWI Korean morphological analyzer so the LLM is invoked only on sentence-ending or symbol cues — avoiding unnecessary calls.
- Cut response latency from ~4 s to under 1 s ($\approx 4\times$) via this lightweight invocation logic plus prompt engineering tuned on a financial-fraud dataset.
- Now re-architecting from a server pipeline to on-device AI to resolve privacy limits, and preparing for commercialization.
- **What I studied:** *real-time LLM serving and latency optimization, domain-specific prompt engineering, and the privacy trade-off between server and on-device inference.*

LLM Agent Research — OpenClaw internals & multi-agent extension

2025 – present

Undergraduate research, AI Lab

- Dissected the TypeScript codebase of the open-source AI agent “OpenClaw,” tracing how the Lane Queue (task scheduling), Agent loop (perceive–decide–act cycle), and Pi-agent-core abstractions mediate user–server–LLM interaction.
- Deployed the agent as a research tool to discover new financial-fraud patterns and auto-draft server defense logic; it autonomously searched the web and scaffolded code, sharply accelerating development.
- Found a critical reliability gap: the agent ignored real-world physical input variables (typos, typing latency on phones), producing insecure logic — which led me to adopt human-in-the-loop (HITL) review instead of trusting agent-generated code blindly.
- Extending the system toward role-specialized agents (e.g., CEO / CTO / code-reviewer) to study multi-agent orchestration and reliability.
- **What I studied:** *agent harness architecture, tool-use reasoning, autonomous code generation, multi-agent orchestration, and trustworthy / human-in-the-loop autonomy.*

Time-Series Forecasting Study — TimeXer on financial data

2025

- Read the TimeXer paper and ran the model on financial time-series, comparing predicted vs. actual values to assess its fit for anomaly- and fraud-style prediction.

AI Vocabulary App — LLM-powered learning service

2025

Planner / Developer · published on the App Store

- Built and shipped a Flutter + Supabase app that uses an LLM to generate vocabulary, quizzes, and podcast-style audio; operated it with real user feedback, experiencing the full idea-to-deployment cycle of an AI product.

“Wrtn Assistant” — 2nd Generative AI Ideathon (with Microsoft)

2024

Team PDAPRO · Top Award

- Proposed a Wrtn portal augmented with an AI-memory feature; reached the finals among 21 universities.

AI STUDY AND COURSEWORK

- **Time-series modeling:** TimeXer (paper study and hands-on runs on financial data); forecasting and anomaly/fraud detection.
- **LLM agents:** code-level analysis of an agent harness (Lane Queue, Agent loop, Pi-agent-core); tool-use and autonomous code generation; multi-agent orchestration with role-based agents; human-in-the-loop reliability.
- **LLM systems engineering:** local / on-device deployment (Gemma 3), prompt engineering, RAG, latency optimization, and FastAPI / WebSocket serving.

- **ML fundamentals:** data preprocessing, one-hot encoding, and ensemble methods (LG Aimers 4th cohort).
- **AI coursework:** Artificial Intelligence, Information Retrieval, Mathematics for AI, Database Systems, Algorithms, and Linear Algebra.

SKILLS AND TECHNIQUES

Programming: Python (primary), TypeScript, C/C++, SQL; Solidity (basic).

AI / ML & LLM: LLM application and local deployment (Gemma 3; OpenAI / Anthropic APIs), prompt engineering, RAG, multi-agent orchestration (LangGraph), agent-harness analysis (OpenClaw), time-series forecasting (TimeXer), PyTorch, ensemble methods, on-device AI.

NLP: KIWI Korean morphological analyzer.

Backend & Data: FastAPI, WebSocket, REST API, relational database design, Supabase.

Certifications: SQL Developer (SQLD), Korea Data Agency (Jun 2025).

Languages: Korean (native), Japanese (JLPT N3, 2025).

Programs: LG Aimers 4th cohort (AI / ML, 2024).

AWARDS AND HONORS

- **Excellence Award**, 12th National ICT Convergence Contest — Chungbuk Institute of Science & Technology Innovation (Nov 2025). For the LLM-based anti-phishing AI keyboard; led backend and DB design.
 - **Grand Prize (대상)**, DB Dream Leader “Dream Labs” Competition — DB Kim Jun-Ki Cultural Foundation (Sep 2025).
 - **Top Award (최우수상)**, 2nd Generative AI Ideathon with Microsoft — Wrtn Technologies & Microsoft (Jul 2024).
 - **Grand Prize (대상)**, SW-Centered University Mileage Scholarship (participation category) — Chungbuk National University (Jul 2024).
 - **DB Dream Leader 2nd-Cohort Scholar**, DB Kim Jun-Ki Cultural Foundation (Feb 2025 – Feb 2027).
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